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Contents

Foreword 5

1 Scope 7

2 Normative references 7

3 Definitions and abbreviations 8
3.1 Definitions 8
3.2 Abbreviations 8

4 Session tests 8
4.1 Session function tests 9
4.2 Document function tests 9
4.3 Application service tests 10

5 Application layer tests 10

Annex A (normative): Session Protocol Data Unit (SPDU) list 11

A.1 Session protocol element 11
Table A.1: < CSS > Session protocol element (Valid cases) 12
Table A.2: < RSSP > Session protocol element (Valid cases) 14
Table A.3: < RSSN > Session protocol element (Valid cases) 17
Table A.4: < CSE > Session protocol element (Valid cases) 20
Table A.5: < RSEP > Session protocol element (Valid cases) 20
Table A.6: < CSA > Session protocol element (Valid cases) 21
Table A.7: < RSAP > Session protocol element (Valid cases) 21
Table A.8: < CSUI > Session protocol element (Valid cases) 22
Table A.9: < RSUI > Session protocol element (Valid cases) 22

A.2 Document protocol element 23
Table A.10: < CDS > Document protocol element (Valid cases) 24
Table A.11: < CDC > Document protocol element (Valid cases) 26
Table A.12: < CDE > Document protocol element (Valid cases) 27
Table A.13: < RDEP > Document protocol element (Valid cases) 27
Table A.14: < CDPB > Document protocol element (Valid cases) 28
Table A.15: < RDPBP > Document protocol element (valid cases) 28
Table A.16: < RDPBN > Document protocol element (Valid cases) 29
Table A.17: < CDCL > Document protocol element (Valid cases) 30
Table A.18: < RDCLP > Document protocol element (Valid cases) 31
Table A.19: < CDUI > Document protocol element (Valid cases) 33
Table A.20: < RDGR > Document protocol element (Valid cases) 33

A.3 Session layer tests 34

Annex B (normative): Application layer Abstract Test Suite 38

B.1 Tester sending, IUT receiving 38
B.1.1 Operational tests 38
B.1.2 Tolerance test 38
B.1.3 List of tests 39

B.2 Tester receiving, IUT sending 44
B.2.1 Operational tests 44
B.2.2 Translation of PDUs (in subclause B.1.2) 44
B.2.3 Realization of PDUs at communications port 44
B.2.4 List of PDUs 45
Table B.1: S_CONreq 45
Table B.2: S_CAPreq_cmd 46
Table B.3: S_DATA_req (data_default) 47
Table B.4: S_DATA_req (data_density) 48
Table B.5: S_DATA_req (data_paper_format) 49
Table B.6: S_DATA_req (data_compression) 50
Table B.7: S_DATA_req (data_combined) 51
Table B.8: S_DATA_req (data_default_value_list) 52
Table B.9: S_DATA_req (data_identifiers) 53
Table B.10: S_DATA_req (data_image_options) 54
Table B.11: S_DATA_req (data_length_txt_unit) 55

Annex C (normative): Protocol Implementation Conformance Statement (PICS) 56

C.1 Introduction 56

C.2 References 56

C.3 PICS proforma contents and structure 56

C.4 Preprinted table contents 56

C.5 Table/item identification 57

C.6 Guidance for completion 57

C.7 Session/Presentation layer PICS 58

C.8 Application layer PICS 63

Annex D (normative): Protocol Implementation Extra Information for Testing (PIXIT) 65

D.1 Introduction 65

D.2 References 65

D.3 PIXIT proforma contents and structure 65

D.4 Table/item identification 65

D.5 Guidance for completion 65

D.6 Session/Presentation layer PIXIT 66

D.7 Application layer PIXIT 68

Annex E (informative): Bibliography 70

History 71

Foreword

This European Telecommunication Standard (ETS) was produced by the Terminal Equipment (TE) Technical Committee of the European Telecommunications Standards Institute (ETSI).

The text provides detailed conformance tests for equipment offering group 4 class 1 functionality and is based upon CCITT Recommendation T.64 [1] and ISO 9646 Parts 1 [2] and 2 [3].

This ETS is also closely related to a series of other ETSs which are given in clause 1 (Scope).

Transposition dates	
Date of latest announcement of this ETS (doa):	31 May 1995
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Date of withdrawal of any conflicting National Standard (dow):	30 November 1995

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1 Scope

This ETS contains the details of the Interconnection Capability Tests (ITC) for equipment offering group 4 class 1 facsimile functionality. It is based upon CCITT Recommendation T.64 [1] and ISO 9646, Parts 1 [2] and 2 [3].

The concept of the Interconnection Capability Tests (ITC) is derived from the range of test types defined in CCITT Recommendation X.290. ITC is a subset of the "capability and behaviour" types of test and focuses on checking the basic intercommunication process.

NOTE: Full conformance testing is for further study.

This ETS is closely related to the following ETSs on group 4 class 1 facsimile equipment.

ETS 300 080 [9]: "Integrated Services Digital Network (ISDN) - ISDN lower layer protocols for telematic terminals".

ETS 300 112: "Integrated Services Digital Network (ISDN) - Facsimile group 4 class 1 equipment on the ISDN - End-to-end protocols".

ETS 300 087: "Integrated Services Digital Network (ISDN) - Facsimile group 4 class 1 equipment on the ISDN - Functional specification of the equipment".

ETS 300 280: "Terminal Equipment (TE) - Facsimile group 4 class 1 equipment on the Integrated Services Digital Network (ISDN) Terminal - testing".

2 Normative references

This ETS incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriated places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] CCITT Recommendation T.64 (1988): "Conformance testing procedure for the teletex Recommendations".

[2] ISO/IEC 9646-1 (1991): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework, Part 1: General concepts".

[3] ISO/IEC 9646-2 (1991): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework, Part 2: Abstract test suite specification".

[4] CCITT Recommendation T.62 (1988): "Control procedures for teletex and Group 4 facsimile service".

[5] ITU-T Recommendation T.503 (1991): "A document application profile for the interchange of group 4 facsimile documents".

[6] ITU-T Recommendation T.521 (1992): "Communication application profile BT0 for document bulk transfer based on the session service (according to the rules defined in T.62 bis)".

[7] CCITT Recommendation T.563 (1988): "Terminal characteristics for group 4 facsimile apparatus".

[8] CCITT Recommendation F.184 (1988): "Operational provisions for the international public facsimile service between subscriber stations with group 4 facsimile machines (telefax 4)".

[9] ETS 300 080 (1992): "Integrated Services Digital Network (ISDN) - ISDN lower layer protocols for telematic terminals".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of this ETS, the definitions given in CCITT/ITU-T Recommendations T.62 [4], T.64 [1], T.503 [5], T.521 [6], T.563 [7] and F.184 [8] apply. However, for the purposes of this ETS group 4 facsimile terminal equipment is referred to as the "facsimile equipment".

3.2 Abbreviations

For the purposes of this ETS, the abbreviations given are those stated in the relevant CCITT/ITU-T Recommendations (T.62 [4], T.64 [1], T.503 [5], T.521 [6], T.563 [7] and F.184 [8]) and ETS 300 280 plus the following:

DTAM	Data Transfer Access Management
ITC	Interconnection Capability Tests
IUT	Implementation Under Test
ISDN	Integrated Services Digital Network
SPDU	Session Protocol Data Unit

4 Session tests

Tests for the lower layers of the protocol, up to and including the Transport layer, are contained in ETS 300 080 [9].

Tests for the session functions are given in subclause 4.1, for the document functions in subclause 4.2 and for the application service functions in subclause 4.3.

4.1 Session function tests

The following tests detailed in CCITT Recommendation T.64 [1], annex C, shall be performed with the following modifications:

the tables of SPDUs for commands and responses:

CSS,
RSSP,
RSSN,
CSE,
RSEP,
CSUI,
RSUI,

shall be replaced by the modified tables given in annex A of this ETS.

The following tests shall not be performed as they are concerned with "non-basic terminal capabilities" which are not appropriate to group 4 class 1 facsimile terminals:

CDN 1/0;
CGN 0/0;
CGN 0/1;
CGN 0/2;
CGN 0/3;
CGN 91/0.

The following test has a typographical error in CCITT Recommendation T.64 [1]:

DE 0/2 calls for the tester to issue CDS I1-I7, this should read CSS I1-I7.

4.2 Document function tests

The tests detailed in CCITT Recommendation T.64 [1], annex C (normative), shall be performed with the following modifications:

the tables of SPDUs for commands and responses:

CDS,
CDCL,
RDCLP,
CDC,
CDE,
RDEP,
CDPB,
RDPBP,
CDUI,

shall be replaced by the modified tables contained in this ETS.

The following tests shall not be performed as they are concerned with "non-basic terminal capabilities" which are not appropriate to group 4 class 1 terminals:

CDN 21/1;
CDN 21/2;
CDN 21/3;
CDN 21/4.

4.3 Application service tests

The tests detailed in CCITT Recommendation T.64 [1], annex D (normative), shall be performed with the following exceptions:

MG 4;
MG 5;
MG 6.

Of tests MD 2 only Part (A) shall be performed and note 2 is not applicable.

MD 3
MD 4
MD 5
MD 6

MD 8
MD 9
MD 10
MD 11

CG 1
CG 2
CG 3
CG 4

CD 1
CD 2
CD 3
CD 4
CD 5

ED 4

ED 7

5 Application layer tests

Annex B (normative) contains the Abstract Test Suite for the application layer.

Annex A (normative): Session Protocol Data Unit (SPDU) list

See CCITT Recommendation T.64 [1] for an introduction to, and explanation of, the terminology used.

In the tables, figures underlined represent number of characters not values.

A.1 Session protocol element

Table	Protocol element	
A.1	CSS	Command Session Start
A.2	RSSP	Response Session Start Positive
A.3	RSSN	Response Session Start Negative
A.4	CSE	Command Session End
A.5	RSEP	Response Session End Positive
A.6	CSA	Command Session Abort
A.7	RSAP	Session protocol element
A.8	CSUI	Command Session User Information
A.9	RSUI	Response Session User Information

Table A.1: < CSS > Session protocol element (Valid cases)

Command Session Start (CSS)		Session reference				NonBasicSession			Servi-ce ID	Inact-ive timer	Session service funct.	Non-basic term capab	Session user data	Priv-ate use	Non-std capab
			Term ID	Date & time	Addit ref		Misc capab	Windo-w							
Tester sends Testcase	C I 0 D L I x x M	G I 0 1 L I x x M	P I 0 A L I 1 8 V 2 4 M M M	P I 0 B L I 1 8 V 1 4 M M M	P I 0 C L I 0 2 V M	G I 0 2 L I x x M M	P I 0 D L I x x V x x M	P I 0 E L I 0 1 V 0 1 M M	P I 0 8 L I 0 1 V 0 1 M M	P I 1 2 L I x x V x x M	P I 1 4 L I 0 2 V 04 39 M	Not used in G4 Cl 1 fax	G I C 1 L I 0 8 PV A4,06 80,01 02,81 01,00 M M	G I E x L I x x M	P I E 8 L I x x V x x M
V1 With all defined parameters	0 D x x	0 1 x x	0 A 1 8 2 4	0 B 0 E 1 4	0 C 0 2 2	0 2 x x	0 D x x x	0 E 0 1 0 1	0 8 0 1 0 1	1 2 x x x	1 4 0 2 04 39		The same SUD is used for all PDUs	E x x x	E 8 x x x
V2 With P12 p. ex.	0 D x x	0 1 x x	0 A 1 8 2 4	0 B 0 E 1 4		0 2 3		0 E 1 1	0 8 1 1	1 2 x x	1 4 0 2 04 39			E x x x	
V3 With P OC and PG 02	0 D x x	0 1 x x	0 A 1 8 2 4	0 B 0 E 1 4	0 C 0 2 2	0 2 x x	0 D x x x	0 E 0 1 0 1	0 8 0 1 0 1						
V4 LI defn on 3 octets	0Fxx DFxx	0Fxx 1Fxx	0F01 AF08 2 4	0F00 BF0E 1 4		0 2 3		0 E 1 1	0 8 1 1						

Table A.1: < CSS > Session protocol element (Valid cases) (concluded)

Command Session Start (CSS)		Session reference				NonBasicSession			Servi-ce ID	Inact-ive timer	Session service funct.	Non-basic term capab	Session user data	Priv-ate use	Non-std capab
Tester sends Testcase	C I 0 D	G I 0 1 M	Term ID P I 1 A M	Date & time P I 1 B M	Addit ref P I V 0 0 C 2 2	G I 0 2 M	Misc capab P I V 0 x 0 D x x	Windo-w P I V 0 0 0 E 1 1 M	P I V 0 0 0 8 1 1 M	P I V 1 x x 2 x x	P I V 1 0 04 4 2 39	Not used in G4 Cl 1 fax	PV G I C 1 M A4,06 80,01 02,81 01,00	G I E x	P I V 8 x x
V5 Undefined P and PLI = 0	0 D	0 1	2 4 A 8	0 0 1 4 B E	0 0 C O (PLI = 0)	0 0 2 6	0 0 0 F 1 1 *	0 0 0 E 1 1 *	0 0 0 8 1 1				The same SUD is used for all PDUs		E 0 8 0
V6 Without options	0 D	0 1	2 4 A 8	0 0 1 4 B E		0 0 2 3		0 0 0 E 1 1	0 0 0 8 1 1						
V7 Separator "-" in PV 0B	0 D	0 1	2 4 A 8	0 0 1 4 B E ****		0 0 2 3		0 0 0 E 1 1	0 0 0 8 1 1						
V8 Wrong PV in service ID	Not appropriate for group 4 class 1 facsimile														

Table A.2: < RSSP > Session protocol element (Valid cases)

Response Session Start Positive (RSSP) / Tester sends / Testcase		Session reference						NonBasicSession			Servi-ce ID	Sessn contr funct	Inact-ive timer	Sessn servic funct	Non-basic term capab	Session user data	Private use		Non-std capab
			Term ID	Date & time	Addit ref		Misc capab	Windo-w											
	R L I I 0 x E x	G L I I 0 x 1 x M M	P L P I I V 0 1 2 9 8 4 M M M	P L P I I V 0 0 1 B E 4 M M M note2	P L P I I V 0 0 2 C 2 2	G L I I 0 x 2 x M M	P L P I I V 0 x x D x x	P L P I I V 0 0 0 E 1 1 M M M	P L P I I V 0 0 0 8 1 1 M M M	P L P I I V 1 x x 0 x x	P L P I I V 1 x x 2 x x	P L P I I V 1 0 x 4 2 x		G L P I I V C 0 x 1 8 x M M M	G L I I E x x x	P L P I I V F x x x x x	P L P I I V E x x 8 x x
V1 With all defined parameters	0 x E x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E	0 0 2 C 2 note1	0 x 2 x	0 x x D x x	0 0 0 E 1 1	0 0 0 8 1 1	1 x x 0 x x	1 x x 2 x x	1 0 04 4 2 39	Not used in Gp 4 Cl 1 fax	A4,06 80,01 02,81 01,00	E x x x	F x x x x x	E x x 8 x x
V2 With all P of CSS	0 x E x	o x 1 x	0 1 2 4 9 8	0 0 1 4 B E	0 0 2 C 2 note1	0 x 2 x	0 x x D x x note1	0 0 0 E 1 1	0 0 0 8 1 1	1 x x 0 x x	1 x x 2 x x note1	1 0 04 4 2 39		All PDUs	E x x x no1	F x x x x x note1	
V3 With more PV than CSS	0 x E x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E	0 0 2 C 2 note1	0 x 2 x	0 x x D x x note1	0 0 F E 1 F *	0 0 0 8 1 1	1 000 0 210 **	1 x x 2 x x note1	1 0 04 4 2 39		use the same SUD	E x x x no1	F x x x x x note1	
V4 With undefined parameters	0 x E x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E	0 0 2 C 2 note1	0 x 4 x *	0 0 9 F 9 x *	0 0 0 E 1 1	0 0 0 8 1 1								

Table A.2: < RSSP > Session protocol element (Valid cases) (continued)

Response Session Start Positive (RSSP) / Tester sends / Testcase	RI 0 E	LI x x	Session reference: GI	LI	Term ID: PI	LI	PV	Date & time: PI	LI	PV	Addit ref: PI	LI	PV	NonBasicSession: GI	LI	Misc capab: PI	LI	PV	Windo-w: PI	LI	PV	Servi-ce ID: PI	LI	PV	Sessn contr funct: PI	LI	PV	Inact-ive timer: PI	LI	PV	Sessn servic funct: PI	LI	PV	Non-basic term capab	Session user data: GI	LI	PV	Private use: GI	LI	PI	LI	PV	Non-std capab: PI	LI	PV
V5 With PV = 0	0	x	0	x	0	1	2 4	0	0	1 4	0	0	2	0	0	0	0	0	0	0	0	0	0	0																					
	E	x	1	x	9	8		B	E		C	2	note1	2	6	D	1	0 *	E	1	1	8	1	1										The same SUD is used for all PDUs									
V6 With PV error	0	x	0	x	0	1	2 4	0	0	1 4	0	0	2	0	0				0	0	0	0	0	0	1	0	0																		
	E	x	1	x	9	8		B	E		C	2	note1	2	3				E	1	1	8	1	1	0	1	5 *																		
V7 LI present in 3 octets	0	x	0	x	0F01		2 4	0F00		1 4	0	0	2	0	0				0F000			0	0	0																					
	E	x	1	x	9F08			BF0E			C	2	note1	2	5				EF011			8	1	1																					
V8 Without options	0	x	0	x	0	1	2 4	0	0	1 4				0	0				0	0	0	0	0	0																					
	E	x	1	x	9	8		B	E					2	3				E	1	1	8	1	1																					

Table A.2: < RSSP > Session protocol element (Valid cases) (concluded)

Response Session Start Positive (RSSP) / Tester sends / Testcase		Session reference				NonBasicSession			Servi-ce ID	Sessn contr funct	Inact-ive timer	Sessn servic funct	Non-basic term capab	Session user data	Private use		Non-std capab
	R L I I 0 x E x	G L I I 0 x 1 x M M	Term ID P L P I I V 0 1 2 9 8 4 M M M	Date & time P L P I I V 0 0 1 B E 4 M M M note2	Addit ref P L P I I V 0 0 2 C 2 2	G L I I 0 x 2 x M M	Misc capab P L P I I V 0 x x D x x	Windo-w P L P I I V 0 0 0 E 1 1 M M M	P L P I I V 0 0 0 8 1 1 M M M	P L P I I V 1 x x 0 x x	P L P I I V 1 x x 2 x x	P L P I I V 1 0 x 4 2 x		G I C 1 M L I 0 8 M P V x x x	G L I E x x	P L P I I V F x x x x x	P L P I I V E x x 8 x x
V9 With PV of TID coded "+++"	0 x E x	0 x 1 x	0 1++ ++ 9 8++ **	0 0 1 4 B E	0 0 2 C 2 note1	0 0 2 3		0 0 0 E 1 1	0 0 0 8 1 1					The same SUD is used for all PDUs			
V10 With LI = 0	0 x E x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E		0 0 2 3		0 0 0 E 1 1	0 0 0 8 1 1						E 0 x 0 *		

NOTE 1: Present if sent in CSS only.

NOTE 2: Equal to parameters in CSS.

Table A.3: < RSSN > Session protocol element (Valid cases)

Response Session Start Negative (RSSN) / Tester sends / Testcase		Session reference							NonBasicSession						Sessn servic funct.	Non-basic term capab	Reason			Session user data			Private use				Non-std capab		
				Term ID			Date & time				Addit ref				Misc capab			Windo -w			Servi -ce ID										
	R I 0	L I x	G I 0	L I x	P I 0	L I 1	P V 2	P I 0	L I 0	P V 1	P I 0	L I 0	P V	G I 0	L I x	P I 0	L I x	P V x	P I 0	L I 0	P V 0	P I 0	L I 0	P V 0	P I 1	L I 0	P V x		P I 3	L I x	P V x	G I C	L I 0	P V x	G I E	L I x	P I F	L I x	P V x	P I E	L I x	P V x
	C	x	1 M	x M	9 M	8 M	4 M	B M	E M	4 M note2	C	2	2	2 M	x M	D	x	x	E M	1 M	1 M	8 M	1 M	1 M	4	2	x		2	x	x	1 M	8 M	x M	x	x	x	x	8	x	x
V1 all defined parameters	0	x	0	x	0	1	2 4	0	0	1 4	0	0	2	0	x	0	x	x	0	0	0	0	0	0	1	0	04	Not used in Gp 4 Cl 1 fax	3	0	0	A4,06 80,01			E	x	F	x	x	E	x	x
	C	x	1	x	9	8		B	E		C	2	note1	2	x	D	x	x	E	1	1	8	1	1	4	2	39		2	1	0	02,81 01,00			x	x	x	x	x	8	x	x
V2 With all P of CSS	0	x	o	x	0	1	2 4	0	0	1 4	0	0	2	0	x	0	x	x	0	0	0	0	0	0	1	0	04		71TA2 34D(69) 262						E	x	F	x	x	E	x	x
	C	x	1	x	9	8		B	E		C	2	note1	2	x no1	D	x	x note1	E	1	1	8	1	1	4	2	39 note1					All PDUs use the same SUD		x no1	x	x	x note1	8	x	x note1
V3 With more PV than in CSS	0	x	0	x	0	1	2 4	0	0	1 4	0	0	2	0	x	0	x	x	0	0	F	0	0	0	1	0	04		3	0	0				E	x	F	x	x	E	x	x
	C	x	1	x	9	8		B	E		C	2	note1	2	x no1	D	x	x note1	E	1	F	8	1	1	4	2	39 note1		2	1	1				x	x no1	x	x note1	8	x	x

Table A.3: < RSSN > Session protocol element (Valid cases) (continued)

Response Session Start Negative (RSSN)		Session reference				NonBasicSession			Servi-ce ID	Sessn servic funct	Non-basic term capab	Reason	Session user data	Private use		Non-std capab
			Term ID	Date & time	Addit ref		Misc capab	Windo-w								
Tester sends	R L I I 0 x	G L I I 0 x	P L P I I V 0 1 2	P L P I I V 0 0 1	P L P I I V 0 0	G L I I 0 x	P L P I I V 0 x x	P L P I I V 0 0 0	P L P I I V 0 0 0	P L P I I V 1 0 x		P L P I I V 3 x x	G L P I I V C 0 x	G L I I E x	P L P I I V F x x	P L P I I V E x x
Testcase	C x	1 x M M	9 8 4 M M M	B E 4 M M M note2	C 2 2	2 x M M	D x x	E 1 1 M M M	8 1 1 M M M	4 2 x		2 x x	1 8 x M M M	x x	x x x	8 x x
V4 With undefn parameters	0 x C x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E	0 0 2 C 2 note1	0 x 4 x *	0 0 9 F 9 x *	0 0 0 E 1 1	0 0 0 8 1 1				All PDUs use the same SUD			
V5 With PV = 0	0 x C x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E	0 0 2 C 2 note1	0 0 2 6	0 0 0 D 1 0 * *	0 0 0 E 1 1	0 0 0 8 1 1							
V6 With PV error	0 x C x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E	0 0 2 C 2 note1	0 0 2 6		0 0 0 E 1 1	0 0 0 8 1 1	1 0 FF 4 2 FF **						

Table A.3: < RSSN > Session protocol element (Valid cases) (concluded)

Response Session Start Negative (RSSN) / Tester sends / Testcase		Session reference				NonBasicSession			Servi-ce ID	Sessn servic funct.	Non-basic term capab	Reason	Session user data	Private use		Non-std capab
			Term ID	Date & time	Addit ref		Misc capab	Windo-w								
	R I 0 x C x	G I 0 x 1 x M	P I 0 1 2 9 8 4 M M M	P I V 0 0 1 B E 4 M M M note2	P I V 0 0 C 2 2	G I 0 x 2 x M	P I V 0 x x D x x	P I V 0 0 0 E 1 1 M M M	P I V 0 0 0 8 1 1 M M M	P I V 1 0 x 4 2 x		P I 3 x x 2 x x	G I C 0 x 1 8 x M M M	G I E x x x	P I V F x x x x x	P I V E x x 8 x x
V7 LI present in 3 octets	0 x C x	0 x 1 x	0F012 4 9F08	0F001 4 BF0E	0 0 2 C 2 note1	0 0 2 5		0F000 EF011	0 0 0 8 1 1				All PDUs use the same SUD			
V8 Without options	0 x C x	0 x 1 x	0 1 2 4 9 8	0 0 1 4 B E		0 0 2 3		0 0 0 E 1 1	0 0 0 8 1 1							
V9 Without parameters	0 0 C 0															
V10 Service ID not 01		Not appropriate for group 4 class 1 facsimile														

RSSN Invalid cases are inappropriate.

NOTE 1: Present if sent in CSS only.

NOTE 2: Equal to parameters in CSS.

Table A.4: < CSE > Session protocol element (Valid cases)

Table A.4: < CSE > Session protocol element (Valid cases)

Command Session End (CSE)		Session termination P			
Tester sends Testcase	CI 09	LI x x	PI 1 1	LI x x	PV x x	Optional
V1 Without parameter	0 9	0 0	-			
V2 With P 11 retain xport	0 9	0 3	1 1	0 1	0 0	
V3 With P 11 but release xport	0 9	0 3	1 1	0 1	0 1	
V4 With PL1 = 0	0 9	0 3	1 1	* 0 0	* (-)	

Table A.5: < RSEP > Session protocol element (Valid cases)

Table A.5: < RSEP > Session protocol element (Valid cases)

Response Session End Positive (RSEP)		
Tester sends Testcase	RI 0A	LI 0 0
V1 Standard	0 A	0 0
V2 Three octet LI	0 A	*** F00 F00

Table A.6: < CSA > Session protocol element (Valid cases)

Table A.6: < CSA > Session protocol element (Valid cases)

Command Session Abort (CSA)		Session termination		
	C I	L I	P I	L I	P V
Tester sends Testcase	1 9	x x	1 1 M	x x M	x x M
V1 With P11	1 9	0 3	1 1	0 1	0 0
V2 Other PV	1 9	0 3	1 1	0 1	0 5
V3 Three octet LI	1 9	0 5	1 1	F00 F01	0 1

Table A.7: < RSAP > Session protocol element (Valid cases)

Table A.7: < RSAP > Session protocol element (Valid cases)

Response Session Abort Positive (RSAP)		
	R I	L I
Tester sends Testcase	1 A	0 0
V1 Normal	1 A	0 0
V2 Three octet LI	1 A	*** F00 F00

Table A.8: < CSUI > Session protocol element (Valid cases)

Table A.8: < CSUI > Session protocol element (Valid cases)

Command Session User Information (CSUI)	C I	L I	Document protocol element (DPE)
Tester sends			X X XXXX
Testcase	0	0	X X XXXX
	1	0	M M M
V1 With DPE	0	0	DPE depends on
	1	0	L6 state
V2 Three octet LI	0	F00	DPE depends on
	1	F00	L6 state

Table A.9: < RSUI > Session protocol element (Valid cases)

Table A.9: < RSUI > Session protocol element (Valid cases)

Response Session User Information (RSUI)	R I	L I	Request session function P I	L I	P V	Document Protocol Element (DPE)
Tester sends						
Testcase	0	0	1	x	x	X X XXXX
	2	0	0	x	x	X X XXXX
						M M M
V1 With parameter and user info	0	0	1	0	0	DPE depends on
	2	3	0	1	1	L6 state
V2 Without parameter	0	0		-		DPE depends on
	2	0				L6 state
V3 Three octet LI	0	0	1	F00	0	DPE depends on
	2	9	0	F01	1	L6 state
V4 with undefined PV	0	0	1	0	0	DPE depends on
	2	3	0	1	5	L6 state

A.2 Document protocol element

Table		Protocol Element
A.10	CDS	Command Document Start
A.11	CDC	Command Document Continue
A.12	CDE	Command Document End
A.13	RDEP	Response Document End Positive
A.14	CDPB	Command Document Page Boundary
A.15	RDPBP	Response Document Page Boundary Positive
A.16	RDPBN	Response Document Page Boundary Negative
A.17	CDCL	Command Document Capability List
A.18	RDCLP	Response Document Capability List Positive
A.19	CDUI	Command Document User Information
A.20	RDGR	Response Document General Reject

Table A.10: < CDS > Document protocol element (Valid cases)

Table A.10: < CDS > Document protocol element (Valid cases)

Command Doc Start (CDS)	Comm. Ident.	Doc Ref No.	Doc type ID	Non-Basic Term Capability	Session User Data	Private Use Param.	
Tester sends Testcase	C I 2 D x L I x	P I 2 9 M x L I x M P V x x M	P I 3 0 x L I 0 1 x P V x x M	Not used in Group4 Class 1 Facsimile	G I C 1 M x L I 0 8 M x P V x x M	G I E x x L I x	P I F x L I x
V1 Optional param only used if negotiated		Not appropriate for group 4 class 1 facsimile.			A4,06 80,01,02, 81,01,00		
V2 Without opt parameters	2 0 D E	2 0 33 9 2 01			The same SUD is used for all PDUs.		
V3 With document ID PV 02 Control Doc		Not appropriate for group 4 class 1 facsimile					
V4 With document ID PV 03 Monitor Doc		Not appropriate for group 4 class 1 facsimile					

Table A.10: < CDS > Document protocol element (Valid cases) (concluded)

Command Doc Start (CDS)	Comm. Ident.	Doc Ref No.	Doc type ID	Non-Basic Term Capability	Session User Data	Private Use Param.
Tester sends Testcase	CI 2D / LI xx	PI 29M / LI xx / PV xxM	PI 30 / LI 01 / PV xx	Not used in Group4 Class 1 Facsimile	GIC1M / LI 08 / PV xxM	GIEx / LI xx / PIFx / LI xx
V5 Normal document for interworking		Not appropriate for group 4 class 1 facsimile				
V6 Control doc for interworking		Not appropriate for group 4 class 1 facsimile				
V7 LI 3 octets	20 DF	2 F00 3 9 F01 2 ***			The same SUD is used for all PDUs	
V8 PLI = 0	20 DC	2 0 3 9 1 1 *				EX 00 *
V9 Operator document		Not appropriate for group 4 class 1 facsimile				

Table A.11: < CDC > Document protocol element (Valid cases)

Command Doc Continue (CDC)		Document Linking							Service interwkg ident	Doc ref No.	Doc type ID	Non-basc term capabilities	Session user data	Priv use param
			called term ID	calling term ID	Date & time	Add ref	Sess No.	Doc ref No	Chekpnt ref No.					
Tester sends Testcase	C L I I 1 x D x	G L I I 2 x 1 x	P L P I I V 0 1 2 9 8 4	P L P I I V 0 1 2 A 8 4	P L P I I V 0 0 1 B E 4	P L P I I V 0 0 2 C 2		P L P I I V 2 x x 9 x x M M M	P L P I I V 2 x x A x x M M M	P L P I I V 2 x x 8 x x	P L P I I V 2 x x 9 x x M M M	P L P I I V 3 0 x 0 1 x		G L P I I V C x x 1 x x	G L P I I V E x x x x x
V1 With all previous document presented parameters	1 x D x	2 x 1 x	0 1 24 9 8	0 1 24 A 8	0 0 14 B E	0 0 2 C 2		2 x x 9 x x	2 x x A x x	Not used in Gp 4 Cl 1 fax	2 0 3 9 1 2		Not used in Gp 4 Cl 1 fax	The SUD shall be identical to the initial CDS SUD.	E x x x x x
V2 Only mandatory parameters (note)	1 x D x	2 0 1 A						2 0 333 9 3 001	2 0 333 A 3 001		2 0 333 9 3 009				

NOTE: This case is valid only if this CDC occurs in the same session as the initial CDS.

Table A.12: < CDE > Document protocol element (Valid cases)

Table A.12: < CDE > Document protocol element (Valid cases)

Command Document End (CDE) Tester sends Testcase	CI 29 LI x0	Checkpoint ref number PI 2A LI x0 PV x0 MM MM
V1 CRN = 1, after CDS or last CRN+1	29 x0	2A x0 333 xxx
V2 Three octet LI	29 x0	2A F00 F03 333 xxx

Table A.13: < RDEP > Document protocol element (Valid cases)

Table A.13: < RDEP > Document protocol element (Valid cases)

Response Document End Positive (RDEP) Tester sends Testcase	RI 2A LI x x	Checkpoint ref number PI 2A LI x x PV 3 x MM MM
V1 CRN length and PV equal to CDE	2A x x	2A x x 3333 xxxx
V2 CRN PV equal to CDE, length not equal	2A x x	2A x x * 3333 xxxx ****
V3 Three octet LI	2A x x	2A F00 F03 333 xxx

Table A.14: < CDPB > Document protocol element (Valid cases)

Table A.14: < CDPB > Document protocol element (Valid cases)

Command Doc. Page Boundary (CDPB)		Checkpoint ref number	
Tester sends Testcase	CI 3 1 / LI 0 x	PI 2 A M / LI 0 x M / PV 0 x M	
V1 CRN = 1 after CDS or last CRN+1	3 0 1 x	2 0 333 A x xxx	
V2 Three octet LI	3 F00 1 F05	2 0 333 A 3 xxx	

Table A.15: < RDPBP > Document protocol element (valid cases)

Table A.15: < RDPBP > Document protocol element (valid cases)

Response Doc. Page Boundary Positive (RDPBP)		Checkpoint ref. number	Receive ability jeopardy
Tester sends Testcase	RI 3 2 / LI x x	PI 2 A M / LI x x M / PV 3 x M	PI 2 E M / LI x x M / PV x x M
V1 CRN length and PV equal to CDPB	3 x 2 x	2 x 3333 A x XXXX	2 0 0 E 1 0
V2 CRN PV equal CDPB, length not equal	3 x 2 x	* **** 2 x 3333 A x 0xxx	2 0 0 E 1 0
V3 Three octet LI	3 x 2 x	*** 2 F00 333 A F03 xxx	2 0 0 E 1 0
V4 Param. 2 E set to 1	3 x 2 x	2 x 333 A x xxx	2 0 0 E 1 1

Table A.16: < RDPBN > Document protocol element (Valid cases)

Table A.16: < RDPBN > Document protocol element (Valid cases)

Response Doc Page Boundary Negative (RDPBN)		Reason	
Tester sends Testcase	RI 30 / LI 0x	PI 32M / LI 01M / PV 0xM	
V1 With param	30 / 03	32 / 01 / 00	
V2 Three octet LI	30 / 05	32 / *** F00 F01 / 0x	

Table A.17: < CDCL > Document protocol element (Valid cases)

Command Doc Capability List (CDCL)		Inactive timer	Storage capacity negotiation	NonBasic Terminal capabilities	Session User Data	Private use parameters	Non-std capabilities
Tester sends (testcase)	C I 3 D L I X X	P I 1 2 L I X X P V X X	P I 2 D L I x x P V xx xx	Not used in Gp4 Cl1 facsimile	G I C 1 L I 0 8 PV see below	G I E x L I x x P V x x P I F x L I x x P V x x	P I E 8 L I x x PV x x
V1 With all defined parameters	3 x D x	1 0 0 2 1 0	2 0 00 D 2 2A		A4,06 80,01,02, 81,01,00	E x x x x x F x x X x x	E x x 8 x x
V2 With unexpected parameters	3 x D x	1 0 0 1 1 0 *	2 0 01 D 2 23	Not used for type approval			
V3 LI defn. on 3 oct.	3 Fxx D Fxx ***	1 F00 0 2 F01 2 ***	2 0 01 D 2 23		The same SUD is used on all PDUs		
V4 With several parameters	3 x D x	1 0 00 2 2 12					
V5 Without parameters	3 0 D A						

Table A.18: < RDCLP > Document protocol element (Valid cases)

Response Doc Capability List Positive (RDCLP) / Tester sends Testcase	RIL	Inactive timer PIL PV	Accept. of CDCL param. PIL PV	Storage capacity negotiatn PIL PV	Non-BasicTerminal capabilities	Session User Data GIL PV	Private use parameters GIL PV / PIL PV	Non-std capab. PIL PV
V1 With all defined parameters	3 x E x	1 0 0 2 1 0	Not used in Gp 4 Class 1 fax	2 0 00 D 2 10		A4,06 80,01,02 81,01,00	E x x / F x x x x x / x x x	E x x 8 x x
V2 With unexpected parameter	3 x E x	1 0 0 1 1 0 *		2 0 01 D 2 23	Not used for type approval			
V3 With un-defn. PV	3 x E x	1 0 0 2 1 0		2 0 0 D 1 4 *	Not used for type approval			
V4 LI defn on 3 octets	3Fxx EFxx ***	1 F00 0 2 F01 2		2 0 0F D 2 8F		The same SUD is used for all PDUs		

Table A.18: < RDCLP > Document protocol element (Valid cases) (concluded)

Response Doc Capability List Positive (RDCLP) / Tester sends Testcase	R I 3 E	L I x x	Inactive timer P I 1 2	L I x x	P V x x	Accept. of CDCL param. P I 2 C	L I x x	P V x x	Storage capacity negotiatn P I 2 D	L I 0 2	P V xx xx	Non-BasicTerminal capabilities Not used in Gp 4 class 1 facsimile	Session User Data G I C 1	L I 0 8	P V see below	Private use parameters G I E x	L I x	P V x	P I F x	L I x	P V x	Non-std capab. P I E 8	L I x x	P V x x
V5 With several parameters	not required for group class 1 facsimile																							
V6 With P 2C PV 01	not required for group 4 class 1 facsimile.																							
V7 With P 2C PV 01 and P	not required for group 4 class 1 facsimile																							
V8 Without parameters	3 E	0 A																						

Table A.19: < CDUI > Document protocol element (Valid cases)

Table A.19: < CDUI > Document protocol element (Valid cases)

Command Document User Information (CDUI)	C I	L I	Correct user information
Tester sends Testcase	0 1	0 0	XXXXXXXXXX XXXXXXXXXX
V1 Normal	0 1	0 0	XXXXXXXXXX XXXXXXXXXX
V2 Three octet LI	0 1	*** F00 F00	XXXXXXXXXX XXXXXXXXXX

Table A.20: < RDGR > Document protocol element (Valid cases)

Table A.20: < RDGR > Document protocol element (Valid cases)

Response Doc General Reject (RDGR)	R I	L I	Reflect param values P I	L I	P V
Tester sends Testcase	0 0	x x	3 1 M	x x M	x x M
V1 With param	0 0	x x	3 1	0 1	x x
V2 With param	0 0	x x	3 1	x x	(x)

A.3 Session layer tests

Implementation Under Test (IUT) calling/Tester called

Test Number	Mandatory	PICS dependent	Additional comments
GN1	Y		
GN2	Y		
GN3	Y		
GN4	Y		RDCLP V2, 3, 5, 6, 7 are not used
GN5	Y		GN5 - GN14 if the IUT
GN6	Y		supports 5 page document
GN7	Y		transmission.
GN8	Y		
GN9	Y		
GN10	Y		
GN11	Y		
GN12	Y		
GN13	Y		
GN14	Y		
GN21	Y		
GN22	Y		
GE7/0	Y		
GE8/0	Y		
GE8/1	Y		
GE8/2	Y		
GE8/3	Y		
GE91/0	Y		
GE91/1	Y		
GE9C/0	Y		
GE93/0	Y		
GE93/1	Y		
GE93/3	Y		
GE93/4	Y		
GE94/0	Y		Ref. to GN tests
GE94/1	Y		should read
GE94/2	Y		GN1 - GN7.
GE94/3	Y		Ref. to GN tests
GE94/4	Y		should read GN1 - GN6.
GE95/0	Y		Ref. to GN tests
GE95/1	Y		should read
GE95/2	Y		GN1 - GN6.

*

IUT calling/Tester called

Test Number	Mandatory	PICS dependent	Additional comments
GE95/4 GE95/5 GE95/6	Y Y Y		Ref. to GN tests should read GN1 - GN7 RDBP I1- I5 not used.
GE96/0 GE96/1 GE96/2		Y Y Y	
GE96/3		Y	
GE97/0 GE97/1	Y Y		
GE9C/0	Y		
GE93/0 GE93/1 GE93/3	Y Y Y		
GE93/4	Y		
GE94/0 GE94/1 GE94/2	Y Y Y		Ref. to GN tests should read GN1 - GN7.
GE94/3 GE94/4	Y Y		Ref. to GN tests should read GN1 - GN6.
GE95/0 GE95/1 GE95/2	Y Y Y		Ref. to GN tests should read GN1 - GN6.
GE95/4 GE95/5 GE95/6	Y Y Y		Ref. to GN tests should read GN1 - GN7 RDBP I1- I5 not used for interconnection capability testing.
GE96/0 GE96/1 GE96/2		Y Y Y	
GE96/3		Y	
GE97/0 GE97/1	Y Y		
E13/0 GE13/1 GE13/2	Y Y Y		

IUT called/Tester calling

Test Number	Mandatory	PICS dependent	Additional comments
GE14/0	Y		
GE14/1	Y		
DN1	Y		CSS V8 is not used.
DN2	Y		
DN4	Y		CSS V8 is not used.
DN5	Y		CDCL V2 is not used.
DN6	Y		
DN7	Y		
DN8	Y		
DN9	Y		
DN10	Y		
DN13	Y		
DN15	Y		
DE0/0	Y		
DE0/1	Y		
DE21/0	Y		
DE21/1	Y		
DE21/2	Y		
DE21/4	Y		
DE21/5	Y		
DE22/0	Y		
DE22/1	Y		
DE22/2	Y		
DE22/3	Y		
DE22/4	Y		
DE22/5	Y		
DE22/6	Y		
DE22/7	Y		
DE22/8	Y		
DE22/9	Y		
DE22/10	Y		
DE22/11	Y		
DE22/12	Y		
DE22/13	Y		
DE23/0	Y		
DE23/2	Y		
DE23/3	Y		
DE23/4	Y		
DE23/10	Y		Inactiv. timer left (CAS=CSA)
DE23/11	Y		at default setting.

IUT called/Tester calling

Test Number	Mandatory	PICS dependent	Additional comments
DE28/0 DE28/1 DE28/2	Y Y Y		
DE29/0 DE29/1 DE29/2	Y Y Y		
DE14/0 DE14/1	Y Y		
MG1 MG2 MG3	Y Y Y		
MD1 MD2 MD7	Y Y Y		note 2 is deleted.
EG1 EG2	Y Y		
ED1 ED2 ED3 ED6	Y Y Y Y		note 2 is deleted.

Annex B (normative): Application layer Abstract Test Suite

The following tests exercise the protocols which are specific to the group 4 class 1 facsimile application of the telematic protocol stack. Only tests 1, 7, 8 and 9 are mandatory, all the others depend on PICS declarations.

B.1 Tester sending, IUT receiving

B.1.1 Operational tests

These tests exercise the normal operation of the terminal in receive mode.

Test 1 - Default
The tester sends document PFTEST (see ETS 300 280) at the default values (200 ppi, A4 format).

Test 2 - Density
The tester sends document PFTEST at all the resolutions stated as managed in the PICS.

Test 3 - Paper_formats
The tester sends document PFTEST with all the paper formats stated as managed in the PICS.

Test 4 - Compression
The tester sends document PFTEST with the "compression" parameter set to "uncompressed mode" if this mode is stated as managed in the PICS.

Test 5 - Combined
The tester sends documents with the combination of all the densities and paper formats stated as managed in the PICS.

B.1.2 Tolerance test

These tests verify the tolerance of the terminal to non-basic features of the application protocol.

Test 6 - Default_value_list
The tester sends document PFTEST at optional transmission density of 240, 300 or 400 ppi, by modifying the default value (200 ppi) by means of the "default value list" parameter.

Test 7 - Identifiers
The tester sends document PFTEST at 200 ppi and uses optional identifiers such as "object identifier", subordinates", "content identifier".

Test 8 - Data_image_options
The tester sends document PFTEST at 200 ppi and uses optional parameters such as "pel path", "line progression", number of pels per line", number of discarded pels", set to their basic values.

Test 9 - Length_txt_unit
The tester sends document PFTEST at 200 ppi and uses all the different ways allowed in ASN.1 (see CCITT Recommendation X.208) to code the length of the content portions of the facsimile document.

B.1.3 List of tests

Test No 1	Type of test	Tester action	Tester detects	SPDUs sent by the tester
	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_1
Default	Send Capability cnf		S_CAPcnf	
	Receive Activity Start	S_ASreq		activity_1
	Receive page	S_DATAreq		data default
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 2	Type of test	Tester action	Tester detects	SPDUs sent by the tester
	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_2
Density	Send Capability cnf		S_CAPcnf	
	Receive Activity Start	S_ASreq		activity_2
	Receive page	S_DATAreq		data density
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 3	Type of test	Tester action	Tester detects	SPDUs sent by the tester
Page format	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_3
	Send Capability cnf		S_CAPcnf	
	Receive Activity Start	S_ASreq		activity_3
	Receive page	S_DATAreq		data_pageformat
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 4	Type of test	Tester action	Tester detects	SPDUs sent by the tester
Compression	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_4
	Send Capability cnf		S_CAPcnf	
	Receive Activity Start	S_ASreq		activity_4
	Receive page	S_DATAreq		data_compression
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 5	Type of test	Tester action	Tester detects	SPDUs sent by the tester
	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_5
Combined	Send Capability cnf		S_CAPcnf	
	Receive Activity Start	S_ASreq		activity_5
	Receive page	S_DATAreq		data_combined
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 6	Type of test	Tester action	Tester detects	SPDUs sent by the tester
	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_2
Default	Send Capability cnf		S_CAPcnf	
value list	Receive Activity Start	S_ASreq		activity_2
	Receive page	S_DATAreq		data_default_value_list
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 7	Type of test	Tester action	Tester detects	SPDUs sent by the tester
	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_1
Identifier	Send Capability cnf		S_CAPcnf	
	Receive Activity Start	S_ASreq		activity_1
	Receive page	S_DATAreq		data_identifier
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 8	Type of test	Tester action	Tester detects	SPDUs sent by the tester
	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_1
Image	Send Capability cnf		S_CAPcnf	
options	Receive Activity Start	S_ASreq		activity_1
	Receive page	S_DATAreq		data_image_options
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

Test No 9	Type of test	Tester action	Tester detects	SPDUs sent by the tester
	Receive Initiate	S_CONreq		connect
	Send Initiate resp		S_CONcnfp	
	Receive Capability	S_CAPreq		activity_1
Length txt	Send Capability cnf		S_CAPcnf	
unit	Receive Activity Start	S_ASreq		activity_1
	Receive page	S_DATAreq		data_length_txt_unit
	Receive Activity end	S_AEreq		
	Send Activity end cnf		S_AEcnf	

B.2 Tester receiving, IUT sending

B.2.1 Operational tests

The previous tests (Numbers 1, 2, 3 and 5) should also be performed in the other direction, "tester receiving", with the optional values as stated in the PICS.

NOTE: Tests 4, 6, 7, 8 and 9 from the IUT are not necessary.

B.2.2 Translation of PDUs (in subclause B.1.2)

S_CONreq	Session Connect Request
S_CONcnfp	Session Connect Confirm Positive
S_CAPreq	Session Capabilities Request
S_CAPcnf	Session Capabilities Confirm
S_ASreq	Session Activity Start Request
S_DATAreq	Session DATA send Request
S_AEreq	Session Activity End Request
S_AEcnf	Session Activity End Confirm

B.2.3 Realization of PDUs at communications port

Tester sending

S_CONreq	CSS (with SUD = 'connect' values)
S_CONcnfp	RSSP (with SUD = 'connect' values)
S_CAPreq	CSUI, CDCL (with SUD = 'activity' values)
S_CAcnf	RSUI, CDS (with SUD = matching set of values)
S_ASreq	CSUI, CDS (with SUD = activity values)
S_DATAreq	+ T.6 string
S_AEreq	CDE
S_AEcnf	RDEP

IUT sending

S_CONreq	CSS (with SUD = implementation dependent)
S_CONcnfp	RSSP (with SUD = 'connect' values)
S_eAPreq	CSUI, CDCL (with SUD = implementation dependent or possibly no CDC for mandatory values
S_CAPcnf	RSUI, RDCLP (with SUD = 'activity' values or no RDCLP if no CDCL received
S_ASreq S_DATAreq	CSUI, CDS (with SUD = matching set of values to 'activity' values) + T.6 string
S_AEreq	CDE
S_AEcnf	RDEP

B.2.4 List of PDUs

See session and document PDU tables for details of other fields in the following commands and responses.

CSS command takes PDU values V6
RSSP command takes values V8
CDCL command takes PDU values V5
RDCLP command takes V8
CDS command takes V2
CDE command takes V1
RDEP command takes V1
CDPB (if used) command takes V1
RDPBP command takes V1
CSE command takes V3
RSEP command takes V1

Table B.1: S_CONreq

Table B.1: S_CONreq

SUD of CSS	Document Characteristics		
		Document Application Profile	Document Archit. Class
Tester sends	P I A 4 L I 0 6	P I 8 0 L I 0 1 P V 0 2 M M M	P I 8 1 L I 0 1 P V 0 0 M M M
connect	A 4 0 6	8 0 0 0 1 2	8 1 0 1 1 0

Table B.2: S_CAPreq_cmd

SUD of CDS	Document Characteristics			Non-Basic Document characteristics					
		Appl. Profile	Archit. Class		Page dimensions	Raster Graph. Coding attrib.		Raster graph. pres. features	
							Compression		Pel transmission density
Tester sends	PI LI A4 06 M M	PI LI PV 90 01 (1) M M M	PI LI PV 81 01 (1) M M M	PI LI A2 LL NM NM	PI LI PI LI PI LI PV PI LI PV A2 LL 30 08 80 02 (2)81 02(2)	PI LI A3 06	PI LI PV 80 01 (1)	PI LI A4 06	PI LI PV 0B 01 (1)
activity_1	A4 06	80 01 02	81 01 00						
activity_3	A4 06	80 01 02	81 01 00	A2 0C	A2 0A 30 08 02 hz 81 02 vt				
activity_2	A4 06	80 01 02	81 01 00	A2 05				A4 03	8B 01 ptd
activity_4	A4 06	80 01 02	81 01 00	A2 05		A3 03	80 01 cp		
activity_5	A4 06	80 01 02	81 01 00	A2 11	A2 0A 30 08 02 hz 81 02 vt			A4 03	8B 01 ptd

ptd = values of the "pel transmission density" parameter.
hz, vt = values for the horizontal and vertical size of the "Page dimensions" parameter.
cp = value of the "Compression" parameter.
npl = value of the "number of pels per line" parameter.
ndp = value of the "number of discarded pels" parameter.

Table B.3: S_DATA_req (data_default)

CSUI/CDUI	Layout object (for document)													
		Object type	Descriptor body											
				Object identifier	Sub-ordinates	Default value list								
							pages attributes							
Tester sends								dimensions	presentation attributes					
									content type	raster graph. attrib.				
											pel_path	line_progression	pel_transm. density	
data_default	A2 03	02 01 00												

CSUI/CDUI	Layout object (for page)												
		Object type	Descriptor body										
				Object identifier	Content portions	dimensions	presentation attributes						
Tester sends								content type	raster graph. attrib.				
										pel_path	line_progression	pel_transm. density	
data_default (suite)	A2 03	02 01 02											

CSUI/CDUI	Content portion								
		Content portion attributes							Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				
						number of pels per line	compression	number of disc. pels	
data_default (suite)	A3 LL								04 (or 24) LL [T.6 string...]

Table B.4: S_DATA_req (data_density)

CSUI/CDUI	Layout object (for document)												
		Object type	Descriptor body										
				Object identifier	Sub-ordinates	Default value list							
							pages attributes						
Tester sends								dimensions	presentation attributes				
									content type	raster graph. attrib.			
											pel_path	line_progression	pel_transm. density
data_density	A2 03	02 01 00											

CSUI/CDUI	Layout object (for page)											
		Object type	Descriptor body									
				Object identifier	Content portions	dimensions	presentation attributes					
Tester sends								content type	raster graph. attrib.			
										pel_path	line_progression	pel_transm. density
data_density (suite)	A2 0C	02 01 02	31 07				A6 05		A103			82 01 ptd

CSUI/CDUI	Content portion								
		Content portion attributes							Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				
						number of pels per line	compression	number of disc. pels	
data_density (suite)	A3 LL								04 (or 24) LL [T.6 string...]

Table B.5: S_DATA_req (data_paper_format)

CSUI/CDUI	Layout object (for document)												
		Object type	Descriptor body										
				Object identifier	Sub-ordinates	Default value list							
							pages attributes						
Tester sends								dimensions	presentation attributes				
									content type	raster graph. attrib.			
											pel_path	line_progression	pel_transm. density
data_paper_format	A2 03	02 01 00											

CSUI/CDUI	Layout object (for page)											
	Object type	Descriptor body										
			Object identifier	Content portions	dimensions	presentation attributes						
Tester sends							content type	raster graph. attrib.				
									pel_path	line_progression	pel_transm. density	
data_paper_format (suite)	A2 0F	02 01 02	31 0A		80 02 hz A4 08 81 02 vt							

CSUI/CDUI	Content portion							
		Content portion attributes						Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes			
						number of pels per line	compression	number of disc. pels
data_paper_format (suite)	A3 LL							04 (or 24) LL (T.6 string...)

Table B.6: S_DATA_req (data_compression)

CSUI/CDUI	Layout object (for document)												
		Object type	Descriptor body										
				Object identifier	Sub-ordinates	Default value list							
							pages attributes						
Tester sends								dimensions	presentation attributes				
									content type	raster graph. attrib.			
											pel_path	line_progression	pel_transm. density
data_compression	A2 03	02 01 00											

CSUI/CDUI	Layout object (for page)											
		Object type	Descriptor body									
				Object identifier	Content portions	dimensions	presentation attributes					
Tester sends								content type	raster graph. attrib.			
										pel_path	line_progression	pel_transm. density
data_compression (suite)	A2 03	02 01 02										

CSUI/CDUI	Content portion								
		Content portion attributes							Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				
						number of pels per line	compression	number of disc. pels	
data_compression (suite)	A3 LL	31 05			A2 03		82 01 cp		04 (or 24) LL [T.6 string...]

Table B.7: S_DATA_req (data_combined)

CSUI/CDUI	Layout object (for document)												
		Object type	Descriptor body										
				Object identifier	Sub-ordinates	Default value list							
							pages attributes						
Tester sends								dimensions	presentation attributes				
									content type	raster graph. attrib.			
											pel_path	line_progression	pel_transm. density
data_combined	A2 03	02 01 00											

CSUI/CDUI	Layout object (for page)											
	Object type	Descriptor body										
			Object identifier	Content portions	dimensions	presentation attributes						
Tester sends							content type	raster graph. attrib.				
									pel_path	line_progression	pel_transm. density	
data_combined (suite)	A2 16	02 01 02	31 11			80 02 hz A4 08 81 02 vt	A6 05	A103			82 01 ptd	

CSUI/CDUI	Content portion								
		Content portion attributes							Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				
						number of pels per line	compression	number of disc. pels	
data_combined (suite)	A3 LL								04 (or 24) LL [T.6 string...]

Table B.8: S_DATA_req (data_default_value_list)

CSUI/CDUI	Layout object (for document)													
		Object type	Descriptor body											
				Object identifier	Sub-ordinates	Default value list								
							pages attributes							
Tester sends								dimensions	presentation attributes					
									content type	raster graph. attrib.				
											pel_path	line_progression	pel_transm. density	
data_default_value_list	A2 0E	02 01 00	31 09			A7 07	A2 05			A103			82 01 ptd	

CSUI/CDUI	Layout object (for page)												
		Object type	Descriptor body										
				Object identifier	Content portions	dimensions	presentation attributes						
Tester sends								content type	raster graph. attrib.				
										pel_path	line_progression	pel_transm. density	
data_default value_list (suite)	A2 03	02 01 02											

CSUI/CDUI	Content portion								
		Content portion attributes							Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				
						number of pels per line	compression	number of disc. pels	
data_default value_list (suite)	A3 LL								04 (or 24) LL [T.6 string...]

Table B.9: S_DATA_req (data_identifiers)

CSUI/CDUI	Layout object (for document)												
		Object type	Descriptor body										
				Object identifier	Sub-ordinates	Default value list							
							pages attributes						
Tester sends								dimensions	presentation attributes				
									content type	raster graph. attrib.			
											pel_path	line_progression	pel_transm. density
data_identifiers	A2 0D	02 01 00	31 08	41 01 31	A003120130								

CSUI/CDUI	Layout object (for page)											
		Object type	Descriptor body									
				Object identifier	Content portions	dimensions	presentation attributes					
Tester sends								content type	raster graph. attrib.			
										pel_path	line_progression	pel_transm. density
data_identifiers (suite)	A2 0F	02 01 02	31 0A	4103312030	A103120130							

CSUI/CDUI	Content portion								
		Content portion attributes							Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				
						number of pels per line	compression	number of disc. pels	
data_identifiers (suite)	A3 LL	31 07	40 05 31 20 30 20 30						04 (or 24) LL [T.6 string...]

Table B.10: S_DATA_req (data_image_options)

CSUI/CDUI	Layout object (for document)													
		Object type	Descriptor body											
				Object identifier	Sub-ordinates	Default value list								
							pages attributes							
Tester sends								dimensions	presentation attributes					
									content type	raster graph. attrib.				
											pel_path	line_progression	pel_transm. density	
data_image_options	A2 03	02 01 00												

CSUI/CDUI	Layout object (for page)												
		Object type	Descriptor body										
				Object identifier	Content portions	dimensions	presentation attributes						
Tester sends								content type	raster graph. attrib.				
										pel_path	line_progression	pel_transm. density	
data_image_options (suite)	A2 12	02 01 02	31 0D				A6 0B	42 01 01	A1 06	80 01 00	81 01 03		

CSUI/CDUI	Content portion								
		Content portion attributes						Content information	
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				
						number of pels per line	compression	number of disc. pels	
data_image_options (suite)	A3 LL	31 0F		80 01 01	A2 0A	82 02 npl	82 01 cp	83 01 ndp	04 (or 24) LL [T.6 string...]

Table B.11: S_DATA_req (data_length_txt_unit)

CSUI/CDUI	Layout object (for document)													
		Object type	Descriptor body											
				Object identifier	Sub-ordinates	Default value list								
							pages attributes							
Tester sends								dimensions	presentation attributes					
									content type	raster graph. attrib.				
											pel_path	line_progression	pel_transm. density	
data_length_ txt_unit	A2 03	02 01 00												

CSUI/CDUI	Layout object (for page)												
		Object type	Descriptor body										
				Object identifier	Content portions	dimensions	presentation attributes						
Tester sends								content type	raster graph. attrib.				
										pel_path	line_progression	pel_transm. density	
data_length_ txt_unit (suite)	A2 03	02 01 02											

CSUI/CDUI	Content portion								
		Content portion attributes							Content information
Tester sends			Content identifier	type of coding	Raster graphics coding attributes				24 80 04 82 LL [T.6 file] or 24 80 04 82 ll [T.6 portion] 04 82 ll [T.6 portion] or 04 82 LL [T.6 file] ... or 24 82 LL 04 82 ll [T.6 port.] 04 82 ll [T.6 portion]
						number of pels per line	compression	number of disc. pels	
data_length_ txt_unit (suite)	A3 80 A3 82LL								

Annex C (normative): Protocol Implementation Conformance Statement (PICS)

C.1 Introduction

To evaluate conformance/approval of a particular implementation, it is necessary to have a statement of which capabilities and options have been implemented for a given protocol. Such a statement is called a Protocol Implementation Conformance Statement (PICS).

A client who requests a conformance/approval test shall provide to the test laboratory a completed PICS proforma for each layer of the equipment to be tested.

This annex comprises the PICS proformas related to end-to-end protocols tests of the B-channels part of ISDN facsimile group 4 class 1 equipment for ISDN basic access and circuit-switched mode (DTE-DTE communication):

Session/Presentation layer;

Application layer.

NOTE: Application layer PICS items related to terminal testing are found in annex D of ETS 300 280.

C.2 References

Session/Presentation layer: CCITT Recommendation T.62 [4].

Application layer: CCITT/ITU-T Recommendation T.432, T.503 [5], T.521 [6], T.563 [7] and F.184 [8].

C.3 PICS proforma contents and structure

The PICS proformas consist of tables structured as indicated in the following documents:

ISO/IEC 9646-1 [2]: "Information technology - Open Systems Interconnection - Conformance testing methodology and framework, Part 1: General concepts".

ISO/IEC 9646-2 [3]: "Information technology - Open Systems Interconnection - Conformance testing methodology and framework, Part 2: Abstract test suite specification".

C.4 Preprinted table contents

The pre-printed contents of the PICS tables provide the following:

- table/item identification, see clause C.5;
- item/facility names or short descriptions;
- references to the Standards;
- status attributes specifying the status of the items;
- columns to be filled in by the client, see clause C.6.

The status attribute in the "STATUS" column reflects the conformance requirements defined in the referenced Standard as follows:

m = mandatory: the item is specified as "mandatory", i.e. the capability is required for approval/conformance.

o = optional: the item is specified as "optional", i.e. the capability is not required for approval/conformance, but if it is implemented it shall conform to the specifications.

However, in order to reduce the time required for type approval testing, some of the status attributes are denoted "testing not applicable for type approval" meaning that these items shall not be tested for Interconnection Capability Testing.

C.5 Table/item identification

The table/item identification in the PICS tables are as follows:

- the table headers contain a label composed of an initial capital letter which is the first letter of the layer name, and a serial (integer) number;

- the rows - each corresponding to an item - are numbered by serial (integer) numbers.

To identify an item, the table label and the item number are combined by use of a slash (solidus), "/", e.g. the third item in the second table in the Session/Presentation layer PICS is identified by "S2/3".

C.6 Guidance for completion

The filling in is done in the right-hand table column named "IMPLEMENTED?". Each row of the column shall be filled in as follows:

- for implemented items a Y (or YES) is entered;

- for not implemented items a N (or NO) is entered.

For each non-implemented mandatory item the client shall provide a justification.

C.7 Session/Presentation layer PICS

C7 S E S S I O N / P R E S E N T A T I O N L A Y E R P I C S

for

end-to-end protocols tests

of

ISDN facsimile group 4 class 1 equipment, B-channels part,
using ISDN basic access and circuit-switched mode (DTE-DTE communication)

REFERENCES
CCITT Recommendation T.62 [4]

TABLE SECTIONS:

S1	MAJOR CAPABILITIES
S2	PDUs regarding SESSION ESTABLISHMENT/CLEARING, ref. S1/1
S3	PDUs regarding SESSION INFORMATION TRANSFER, ref. S1/2
S4	PDUs regarding SESSION MANAGEMENT, ref. S1/3
S5	PDUs regarding DOCUMENT CONTROL, ref. S1/4
S6	PDUs regarding DOCUMENT INFORMATION TRANSFER, ref. S1/5
S7	PDUs regarding ERROR RECOVERY, ref. S1/6
S8	NEGOTIATION at session initiation (CSS/RSSP), ref. S1/1
S9	NEGOTIATION after session initiation (CDCL/RDCLP), ref. S1/4
S10	PARTICULAR PROTOCOL PROCEDURES
S11	PROCEDURES AND REACTIONS IN CASE OF PROTOCOL ERRORS
S12	TIMERS

S1	MAJOR CAPABILITIES				
Item No.	MAIN ITEM	DETAILS IN TABLE	REFERENCE TO CCITT Rec.T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	Session establishment/clearing	S2	Table 1	m	
2	Session information transfer	S3	Table 1	m	
3	Session management	S4	Table 1	o *)	
4	Document information transfer	S5	Table 2	m	
5	Document control	S6	Table 2	m	
6	Document error recovery	S7	Table 2	m	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.					

S2	PDUs regarding SESSION ESTABLISHMENT/CLEARING, ref. S1/1			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	CSS	3.2.1	m	
2	RSSP	3.2.2	m	
3	RSSN	3.2.3	m	
4	CSE	3.2.4	m	
5	RSEP	3.2.5	m	
6	CSA	3.2.6	m	
7	RSAP	3.2.7	m	

S3	PDUs regarding SESSION INFORMATION TRANSFER, ref. S1/2			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	CSUI	3.2.8	m	
2	RSUI	3.2.9	m	

S4	PDUs regarding SESSION MANAGEMENT, ref. S1/3			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	CSCC	3.2.10	o *)	
2	RSCCP	3.2.11	o *)	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.				

S5	PDUs regarding DOCUMENT INFORMATION TRANSFER, ref. S1/4			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	CDUI	3.4.12	m	

S6	PDUs regarding DOCUMENT CONTROL, ref. S1/5			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	CDS	3.4.1	m	
2	CDC	3.4.3	o *)	
3	CDCL	3.4.4	m	
4	RDCLP	3.4.5	m	
5	CDE	3.4.6	m	
6	RDEP	3.4.7	m	
7	CDD	3.4.8	o *)	
8	RDDP	3.4.9	m	
9	CDR	3.4.10	o *)	
10	RDRP	3.4.11	m	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.				

S7	PDUs regarding DOCUMENT ERROR RECOVERY, ref. S1/6			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	RDGR	3.4.2	o *)	
2	CDPB	3.4.13	m	
3	RDPBP	3.4.14	m	
4	RDPBN	3.4.15	m	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.				

S8	NEGOTIATION at session initiation (CSS/RSSP), ref. S1/1			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	Non-basic window size	3.3.2.7, 4.3.2 5.7.2.6	m	
2	Value of Inactivity timer (see table S12)	3.2.2.2 h) 5.7.2.11	o *)	
3	Session service functions	3.2.1.2 h), 5.7.2.12	o *)	
4	Non-standardized capabilities	3.2.2.2 k), 5.7.2.13	o *)	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.				

S9	NEGOTIATION after session initiation (CDCL/RDCLP), ref. S1/4			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	Storage capacity using CDCL	3.4.4.2, 3.4.5.3	o *)	
2	Storage capacity using RDCLP		o *)	
3	Inactivity timer value, see table S12 using CDCL	3.4.4.5, 3.4.5.4	o *)	
4	Inactivity timer value, see table S12 using RDCLP		o *)	
5	Session user data using CDCL	3.4.4.6, 3.4.5.4	m	
6	Session user data using RDCLP		m	
7	Private use using CDCL	3.4.4.6, 3.4.5.4	o *)	
8	Private use using RDCLP		o *)	
9	Non-standardized capabilities using CDCL	3.4.4.7, 3.4.5.4	o *)	
10	Non-standardized capabilities using RDCLP		o *)	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.				

S10	PARTICULAR PROTOCOL PROCEDURES			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	Use of session user data field	3.2.1.2 i), 3.4.1.2 e) 3.7.2.14	m	
2	Release of transport connection if timeout (CSA timer or Inactivity timer)	3.3.2.6 b)	m	
3	Response to CDCL without NBTC	3.4.5	m	
4	Indication of reason code "unable to continue the session" in RDPBN when a sink terminal sends RDPBP with RAJ=1 and subsequent memory overflow results in sending RDPBN	3.5.8	m *)	
5	Handling of document and checkpoint reference numbers	4.2	m	
6	Stop sending when window limit is reached	4.3, Fig.G-3	m	
7	Reception of CDE when outstanding acknowledgement exists	4.3, Fig.G-4	m	
8	Reception of a document page	5.7.2.12	m	
9	Sending of a complete document	5.7.2.12	m	
10	Use of session termination parameter in CSE indicating clearing of the transport connection	3.2.4.1	o *)	
11	Use of acknowledgement change request function	3.2.11, 5.7.2.8	o *)	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.				

S11	PROCEDURES AND REACTIONS IN CASE OF PROTOCOL ERRORS			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	IMPLEMENTED? (Y=Yes/N=No)
	Rejecting a document page by sending:			
1	- CSA	Fig.G-2: State 14	o *)	1)
2	- RDGR	Fig.G-4: State DR 7	o *)	1)
3	- RDPBN	Fig.G-4: State DR 7	o *)	1)
	Responding a document reject (RDGR/RDPBN) by sending:			
4	- CSA	Fig.G-1: State 14	o	1)
5	- CDD	Fig.G-3: State DS 9	o	1)
6	- CDR	Fig.G-3: State DS 8	o	1)
	Responding invalid/unexpected PDUs received during data transfer phase by sending:			
7	- CSA	Fig.G-1: State 14	o *)	1)
8	- CDD	Fig.G-3: State DS 9	o *)	1)
9	- CDR	Fig.G-3: State DS 8	o *)	1)
10	Session termination parameter in CSA	Table 9, 5.7.2.9	m	
11	Reason parameter in RSSN	3.2.3.2 g), 5.7.2.10	o *)	
12	Reason parameter in RSSN provided as a text message	3.2.3.2 g)	o *)	
13	Reason parameter in CDD and/or CDR	3.4.8.2, 3.4.10.2, 5.7.2.10	o *)	
1) Within the items 1-3, 4-6 and 7-9 at least one of the 3 options shall be implemented.				
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.				

S12	TIMERS				
Item No.	ITEM	REFERENCE TO CCITT Rec. T.62	STATUS	RANGES/VALUES	
				ALLOWED	SUPPORTED
1	Basic inactivity timer	4.1.2, 5.7.2.11	m	60 seconds	
2	Basic demand response timer	G.2.2.7 b)	m	60 seconds	
4	Negotiation of Inactivity timer value	4.1.2, 5.7.2.11	o *)	1-63 seconds	
5			o *)	1-63 minutes	
6			o *)	1-63 hours	
*) TESTING NOT APPLICABLE FOR TYPE APPROVAL.					

C.8 Application layer PICS

C8 APPLICATION LAYER PICS

for

end-to-end protocols tests

of

ISDN facsimile group 4 class 1 equipment, B-channels part,
using ISDN basic access and circuit-switched mode (DTE-DTE communication)

REFERENCES
CCITT/ITU-T Recommendation T.503 [5], T.521 [6], T.563 [7], F.184 [8]

TABLE SECTIONS:
- A1 TERMINAL CLASS
- A2 DOCUMENT APPLICATION PROFILE
- A3 COMMUNICATION APPLICATION PROFILE

NOTE: Application layer PICS items related to Terminal Testing are found in annex D of ETS 300 280.

A1	TERMINAL CLASS			
Item No.	ITEM	REFERENCE TO x) CCITT Rec. T.563 y) CCITT Rec. F.184	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	Class 1 operation	x) 1.5, 3.1.6, 3.2.8 y) 1.2.1	m	

A2	DOCUMENT APPLICATION PROFILE			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.503	STATUS	IMPLEMENTED? (Y=Yes/N=No)
1	A document is a succession of pages containing raster-graphics content architecture	5.3.1	m	
2	Paper formats as defined in T.563 [7]	5.3.2	m	
3	Use of vertical page orientation only	5.3.2	m	
4	Raster-graphics content as defined in T.563 [7]	5.4.1	m	

A3	COMMUNICATION APPLICATION PROFILE			
Item No.	ITEM	REFERENCE TO CCITT Rec. T.521	STATUS	IMPLEMENTED? (Y=Yes/N=No)
	Functional units as defined in T.432 for BT0:	5.2		
1	- Association use control (Kernel)		m	
2	- Capability		m	
3	- Document bulk transfer		m	
4	- Token control		m	
5	- Exception report		m	
4	- Reliable transfer mode 1		m	

Annex D (normative): Protocol Implementation Extra Information for Testing (PIXIT)

PIXIT

for

end-to-end protocols tests

of

ISDN facsimile group 4 class I equipment, B-channels part,

using ISDN basic access and circuit-switched mode (DTE-DTE communication)

D.1 Introduction

In order to test a protocol implementation, test related information in addition to that provided by the PICS may be needed. Such extra information is called a Protocol Implementation eXtra Information for Testing (PIXIT).

A client who requests a conformance/approval test shall provide to the test laboratory a completed PIXIT proforma for each layer of the equipment to be tested.

This annex comprises the PIXIT proformas for the Session/Presentation layer and the Application layer related to end-to-end protocols tests of the B-channels part of ISDN facsimile group 4 class I equipment for ISDN basic access and circuit-switched mode (DTE-DTE communication).

D.2 References

Items in the corresponding PICS document (see annex C).

D.3 PIXIT proforma contents and structure

The PIXIT proformas consist of tables for the provision of the extra information - additional to the PICS information - needed for the testing, see the following documents ISO/IEC 9646-1 [2] and ISO/IEC 9646-5.

D.4 Table/item identification

The table/item identification in the PIXIT tables are as follows:

- the table headers contain a label composed of 2 capital letters and a serial (integer) number. The initial capital letter is the first letter of the layer name. The second capital letter is an X (for PIXIT, in order to distinguish between PICS and PIXIT identifications);

- the rows - each corresponding to an item - are numbered by serial (integer) numbers.

To identify an item, the table label and the item number are combined by use of a slash (solidus), "/", e.g. the third item in the first table of the Application layer PIXIT is identified by "AX1/3".

D.5 Guidance for completion

The pre-printed text in the proformas indicates the kind of information to be filled in by the client.

The information provided in a PIXIT shall not conflict with information provided in the corresponding PICS or with the requirements of the Standards.

D.6 Session/Presentation layer PIXIT

S E S S I O N / P R E S E N T A T I O N L A Y E R P I X I T

for

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REFERENCES: PICS items (see annex C).

TABLE SECTION: SX1 CLIENT's ADDITIONAL INFORMATION FOR TESTING

SESSION/PRESENTATION LAYER PIXIT

SX1		CLIENT's ADDITIONAL INFORMATION FOR TESTING
Item No.	Reference to PICS	Facilities/abilities, values/ranges, statements, clarifications etc.
1	S8/1	Non-basic window sizes:
2	S8/4 S9/9 S9/10	Non-standardized capabilities:
3	S9/7 S9/8	Private use:

D.7 Application layer PIXIT

A P P L I C A T I O N L A Y E R P I X I T

for

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of

ISDN facsimile group 4 class I equipment, B-channels part,
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REFERENCES: PICS items (see annex C).

TABLE SECTIONS: AX1 MEANS OF CONTROL/OBSERVATION, TEST COORDINATION PROCEDURES, etc.

AX2 CLIENT's ADDITIONAL INFORMATION FOR TESTING.

APPLICATION LAYER PIXIT

AX1	MEANS OF CONTROL/OBSERVATION, TEST COORDINATION PROCEDURES, etc.	
Item No.	ITEM	ITEM SUPPORTED? (Yes=Y/No=N)
1	Called terminal set-up and release: D_INITIATE_cnf + service primitive (with result parameter = accept) is visible to the user	
2	In case of problems, the user can initiate abort procedure (D_U_ABORT_req)	
3	The user can specify the parameters of the non-basic document characteristic parameter (in D_CAPABILITY_req)	
4	Receiving of D_TOKEN_PLEASE_ind (polling): - Sending of D_CONTROL_GIVE_req is an automatic reaction	
5	- D_TOKEN_PLEASE_ind is ignored	
6	D_CONTROL_GIVE_req can be sent independent of received D_TOKEN_PLEASE_ind so that the other side can send a document	
7	DTAM Protocol terminal: Reaction on invalid message (refer to annex B.3.1 of T.433): - Appropriate internal terminal event occurs	
8	- Issuing both D-PAind outgoing event (to its DTAMSE-user and DAB outgoing event (to its peer DTAMPM)	
9	Sending of a document (via D_TR_req) after establishment of the connection do not require an explicit action of the operator	

APPLICATION LAYER PIXIT

AX2		CLIENT's ADDITIONAL INFORMATION FOR TESTING
Item No.	Reference to PICS	Facilities/abilities, values/ranges, statements, clarifications etc.

Annex E (informative): Bibliography

1) CCITT Recommendation X.290 (1992): "OSI conformance testing methodology and framework for protocol Recommendations for CCITT applications - General concepts".

2) CCITT Recommendation T.432 (1988): "Document transfer and manipulation (DTAM) - Services and protocols - Service definition".

3) ISO/IEC 9646-5 (1991): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 5: Requirements on test laboratories and clients for the conformance assessment process".

4) ETS 300 280: "Terminal Equipment (TE) - Facsimile group 4 class 1 equipment on the Integrated Services Digital Network (ISDN) - Terminal testing".

5) CCITT Recommendation X.208 (1988): "Specification of Abstract Syntax Notation One (ASN.1)".

History

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